

Mohammad Shahid

CFD / Numerical Software Engineer | Computational Modelling | CUDA/LBM

Den Haag, Netherlands | Open to relocation within NL |

mechshahid@gmail.com | +31 651585410

[LinkedIn](#) | [GitHub Portfolio](#)

Profile

Computational Modelling Engineer with experience in CFD, thermo-fluid simulation, multiphysics modelling, and numerical solver verification. Strong background in first-principles modelling, benchmark validation, sensitivity studies, and translating engineering physics into robust simulation workflows. Recently developed CUDA/C++ Lattice Boltzmann Method solvers for Couette flow and lid-driven cavity flow, including GPU implementation, CPU/GPU benchmarking, and comparison with classical CFD benchmark data. Interested in CFD solver development, numerical methods, GPU acceleration, and high-performance engineering simulation software.

Core Competencies

- CFD solver verification, benchmark validation, convergence studies, and numerical robustness assessment
- Numerical methods for engineering simulation: FDM, FVM, FEM fundamentals, LBM, ADI schemes, iterative solvers
- CUDA/C++ implementation of structured-grid solvers and CPU/GPU performance comparison
- Thermo-fluid modelling: conduction, convection, radiation, flow-thermal coupling, and reduced-order thermal models
- Multiphysics modelling: electro-thermal, thermo-mechanical, and EM-thermal simulation workflows
- Scientific programming, simulation automation, and post-processing using C++, CUDA, Python, MATLAB, and Fortran

Selected Numerical Software Projects

CUDA-Accelerated Lattice Boltzmann Solver for Lid-Driven Cavity Flow

[GitHub](#)

- Implemented a CUDA-accelerated D2Q9-BGK Lattice Boltzmann Method solver for incompressible lid-driven cavity flow, including collision, streaming, bounce-back wall treatment, moving-lid boundary conditions, and macroscopic field extraction.
- Validated the $Re=100$ cavity solution against the classical Ghia centreline velocity benchmark and generated streamline, velocity, and benchmark comparison plots using Python.
- Developed a matching serial C++ implementation and performed CPU/GPU performance comparison on a 101×101 grid for 20,000 timesteps, obtaining approximately $4.2\times$ CUDA speedup over the CPU version.

CUDA-Accelerated LBM Couette Flow Benchmark

[GitHub](#)

- Built a CUDA D2Q9-BGK LBM solver for Couette flow with moving-wall bounce-back treatment and analytical velocity-profile validation.
- Compared GPU and serial CPU implementations for identical grids, timesteps, relaxation parameters, and boundary conditions to quantify CUDA acceleration.
- Documented solver structure, validation plots, benchmark timing, and reproducible compilation/run instructions in a GitHub-ready project format.

Finite-Volume and CFD Solver Portfolio

[GitHub Portfolio](#)

- Developed educational CFD solvers for shock-tube flow, convection-diffusion, ADI heat conduction, quasi-1D nozzle flow, incompressible channel flow, lid-driven cavity flow, and Couette flow using Fortran, C++, CUDA, and Python post-processing.
- Implemented core numerical building blocks including explicit/implicit time integration, finite-volume flux evaluation, iterative solvers, boundary-condition handling, grid-based field extraction, and automated plotting.

Work Experience

Guest Researcher – Computational Modelling

TU Eindhoven, Netherlands | 2025–Present

- Refined reduced-order thermal models for 5G/6G active antenna units to improve front- and rear-side heat-dissipation estimates under realistic convection and radiation constraints.
- Checked simplified predictions against 3D CST simulations across geometry and emissivity variations to tighten screening accuracy and define model validity limits.

Researcher – Computational Modelling**TU Eindhoven, Netherlands | 2020–2025**

- Developed compact analytical and numerical models for thermal management and electro-thermal behaviour in RF hardware, enabling faster early-stage assessment than full 3D simulation alone.
- Evaluated passive and assisted cooling concepts for full-scale 5G/6G active antenna units to clarify feasibility limits, design trade-offs, and thermal operating windows.
- Quantified sensitivity to convection, emissivity, radiation exchange, geometry, and interface assumptions to separate robust conclusions from setup-dependent results.
- Verified simplified models with 3D CST simulations and converted results into operating limits, validation criteria, and practical engineering guidelines.

Early Stage Researcher – Multiphysics Modelling, EU MyWave**TU Eindhoven, Netherlands | 2020–2023**

- Developed first-principles electro-thermal models for passive intermodulation and localized self-heating in high-frequency structures.
- Built coupled CST workflows linking EM loss, temperature rise, deformation, and RF response for thermally loaded RF hardware.
- Derived reduced-order thermal models for electro-thermal PIM structures and checked them against high-fidelity numerical results to create faster assessment workflows.

R&D Intern – Multiphysics Modelling**Ericsson, Sweden | 2021–2022**

- Developed a compact MATLAB thermal resistance model for natural heat dissipation in active antenna units and benchmarked it against CST simulations to assess where simplified screening remained reliable.
- Built sequential electro-thermal-mechanical CST workflows to assess how thermal loading and deformation could affect RF performance.
- Clarified modelling limits, sensitivity drivers, and validation needs before cooling-design decisions, keeping conclusions aligned with available numerical evidence.

M.Sc Intern – Flow-Thermal and Plasma Solver Validation**Akryvia, France | 2019**

- Verified and validated the in-house YAPS plasma-flow solver by testing geometry treatment, boundary conditions, numerical settings, and benchmark cases.
- Performed convergence and robustness studies involving mesh refinement, under-relaxation, and progression-factor sensitivity.
- Validated canonical flow cases including Poiseuille flow, de Laval nozzle flow, laminar round jet, and free-burning argon arc against analytical, numerical, and experimental references.

M.Tech Thesis – Unsteady Heat-Transfer and Flow Simulation

- Developed and adapted an in-house finite-difference Fortran solver for unsteady mixed-convection heat-transfer simulations using body fitted mesh around a torsionally oscillating heated circular cylinder.
- Performed grid-independence checks, time-step sensitivity studies, and benchmark validation against published data before analysing wake behaviour, force response, and Nusselt-number trends.

Education

-
- **M.Sc. Mechanical Engineering (Computational Mechanics)**, Ecole Centrale de Nantes, France – 2019
 - **M.Tech. Mechanical Engineering (Thermal Sciences)**, Aligarh Muslim University, India – 2018
 - **B.Tech. Mechanical Engineering**, Aligarh Muslim University, India – 2011

Technical Language Skills

-
- **Programming Scientific Computing:** C++, CUDA C/C++, Python, MATLAB, Fortran
 - **CFD Numerical Methods:** Lattice Boltzmann Method (D2Q9-BGK), FDM, FVM, FEM fundamentals, ADI schemes, iterative solvers, boundary-condition implementation, verification and validation
 - **Simulation Tools:** OpenFOAM, ANSYS Fluent, STAR-CCM+, CST Studio Suite, COMSOL
 - **GPU / Performance:** CUDA kernel implementation, CPU/GPU benchmarking, performance comparison, structured-grid solver acceleration
 - **Post-processing Automation:** Python, NumPy, pandas, Matplotlib, ParaView, Excel, L^AT_EX
 - **Engineering Domains:** CFD, heat transfer, thermo-fluid modelling, electro-thermal and thermo-mechanical coupling, reduced-order modelling
 - **Version Control Documentation:** GitHub project structuring, README/report writing, reproducible simulation workflows
 - **Languages:** English (professional), Dutch (A2+), French (A1), Hindi/Urdu (native)